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AQA A Level Further Mathematics (7367)

Topic DA: Graphs

Sub topic DA2: Graph Properties (Eulerian/Hamiltonian)

Topic Questions

AQA A Level Further Mathematics (7367)

Topic Question

Topic	**DA: Graphs**
Sub topic	**DA2: Graph Properties (Eulerian/Hamiltonian)**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

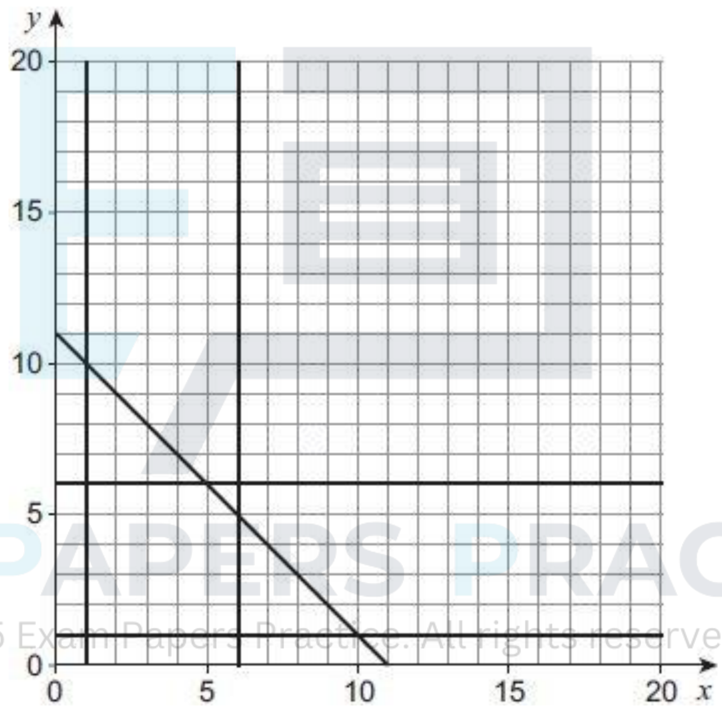
Q1.

A linear programming problem has the constraints

$$1 \le x \le 6$$
$$1 \le y \le 6$$
$$y \ge x$$
$$x + y \le 11$$

(a) (i) Complete **Figure 1** to identify the feasible region for the problem.

Figure 1



(2)

(ii) Determine the maximum value of $5x + 4y$ subject to the constraints.

(2)

(b) The simple-connected graph G has seven vertices.

The vertices of G have degree 1, 2, 3, v, w, x and y

(i) Explain why $x \ge 1$ and $y \ge 1$

(1)

(ii) Explain why $x \le 6$ and $y \le 6$

(1)

(iii) Explain why $x + y \le 11$.

(2)

(iv) State an additional constraint that applies to the values of x and y in this context.

(1)

(c) The graph G also has eight edges. The inequalities used in part (a)(i) apply to the graph G.

(i) Given that $v + w = 4$, find all the feasible values of x and y.

(3)

(ii) It is also given that the graph G is semi-Eulerian.

On **Figure 2**, draw G.

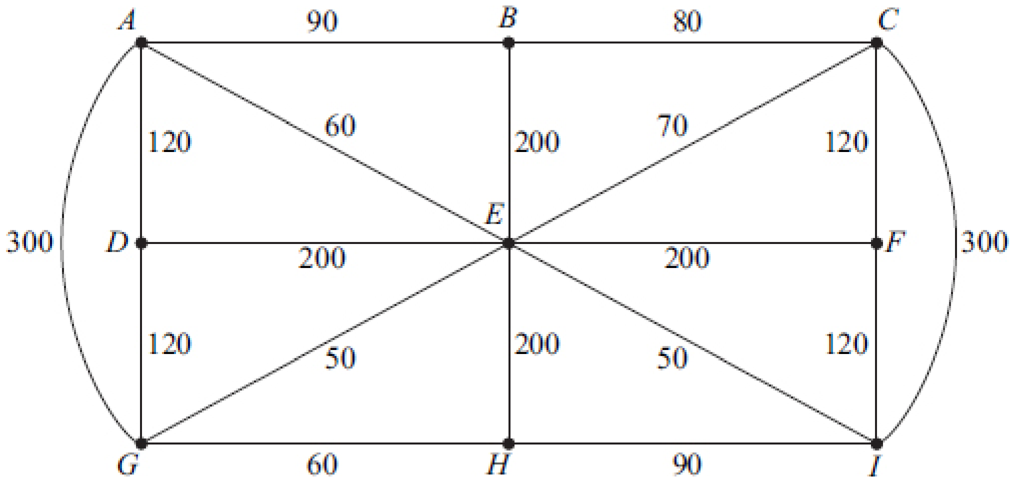
Figure 2

(2)
(Total 14 marks)

Q2.

The network below shows some streets in a town. The number on each edge shows the length of that street, in metres.

Leaflets are to be distributed by a restaurant owner, Tony, from his restaurant located at vertex B. Tony must start from his restaurant, walk along all the streets at least once, before returning to his restaurant.



The total length of the streets is 2430 metres.

(a) Find the length of an optimal Chinese postman route for Tony.

(5)

(b) Colin also wishes to distribute some leaflets. He starts from his house at H, walks along all the streets at least once, before finishing at the restaurant at B.

Colin wishes to walk the minimum distance. Find the length of an optimal route for Colin.

(1)

(c) David also walks along all the streets at least once. He can start at any vertex and finish at any vertex. David also wishes to walk the minimum distance.

(i) Find the length of an optimal route for David.

(1)

(ii) State the vertices from which David could start in order to achieve this optimal route.

(1)

(Total 8 marks)

Q3.

The complete graph K_n ($n > 1$) has every one of its n vertices connected to each of the other vertices by a single edge.

(a) Draw the complete graph K_4.

(1)

(b) (i) Find the total number of edges for the graph K_8.

(ii) Give a reason why K_8 is not Eulerian.

(2)

(c) For the graph K_n, state in terms of n:

(i) the total number of edges;

(ii) the number of edges in a minimum spanning tree;

(iii) the condition for K_n to be Eulerian;

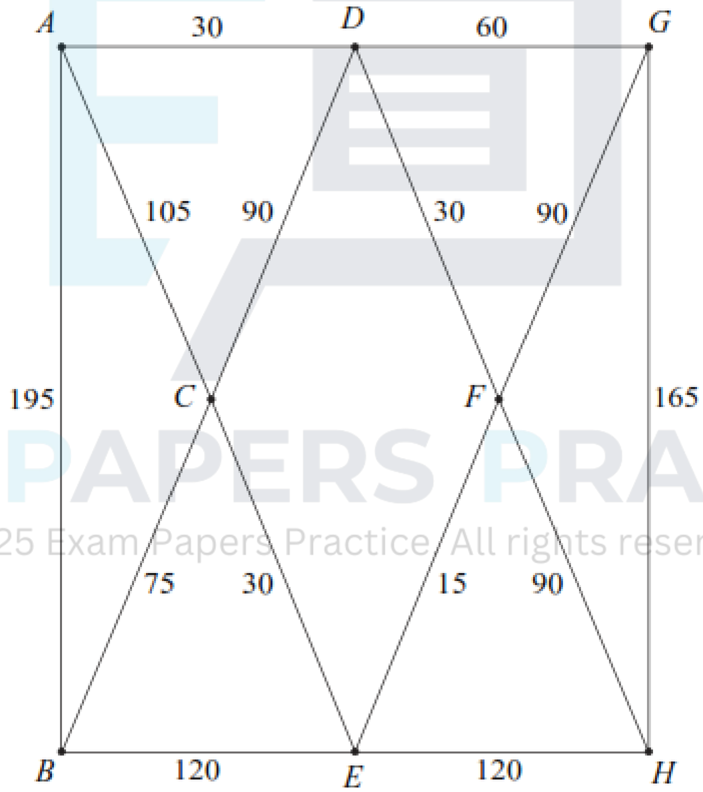
(iv) the condition for the number of edges of a Hamiltonian cycle to be equal to the number of edges of an Eulerian cycle.

(4)
(Total 7 marks)

Q4.

Norris delivers newspapers to houses on an estate. The network shows the streets on the estate. The number on each edge shows the length of the street, in metres.

Norris starts from the newsagents located at vertex A, and he must walk along all the streets at least once before returning to the newsagents.



The total length of the streets is 1215 metres.

(a) Give a reason why it is not possible to start at A, walk along each street once only, and return to A.

(1)

(b) Find the length of an optimal Chinese postman route around the estate, starting and finishing at A.

(5)

(c) For an optimal Chinese postman route, state:

(i) the number of times that the vertex F would occur;

(1)

(ii) the number of times that the vertex H would occur.

(1)
(Total 8 marks)

Q5.

(a) The complete graph K_n has every one of its n vertices connected to each of the other vertices by a single edge.

(i) Find the total number of edges in the graph K_5.

(1)

(ii) State the number of edges in a minimum spanning tree for the graph K_5.

(1)

(iii) State the number of edges in a Hamiltonian cycle for the graph K_5.

(1)

(b) A simple graph G has six vertices and nine edges, and G is Eulerian. Draw a sketch to show a possible graph G.

(2)
(Total 5 marks)

Q6.

A simple connected graph has six vertices.

(a) One vertex has degree x. State the greatest and least possible values of x.

(2)

(b) The six vertices have degrees

$$x - 2,\ x - 2,\ x,\ 2x - 4,\ 2x - 4,\ 4x - 12$$

Find the value of x, justifying your answer.

(2)
(Total 4 marks)

Q7.

A connected graph G has five vertices and has eight edges with lengths 8, 10, 10, 11, 13, 17, 17 and 18.

(a) Find the minimum length of a minimum spanning tree for G.

(2)

(b) Find the maximum length of a minimum spanning tree for G.

(2)

(c) Draw a sketch to show a possible graph G when the length of the minimum spanning tree is 53.

(3)

(Total 7 marks)

Q8.

Angelo is visiting six famous places in Palermo: A, B, C, D, E and F. He intends to travel from one place to the next until he has visited all of the places before returning to his starting place. Due to the traffic system, the time taken to travel between two places may be different dependent on the direction travelled.

The table shows the times, in minutes, taken to travel between the six places.

From \ To	A	B	C	D	E	F
A	–	25	20	20	27	25
B	15	–	10	11	15	30
C	5	30	–	15	20	19
D	20	25	15	–	25	10
E	10	20	7	15	–	15
F	25	35	29	20	30	–

(a) Give an example of a Hamiltonian cycle in this context.

(2)

(b) (i) Show that, if the nearest neighbour algorithm starting from F is used, the total travelling time for Angelo would be 95 minutes.

(3)

(ii) Explain why your answer to part (b)(i) is an upper bound for the minimum travelling time for Angelo.

(2)

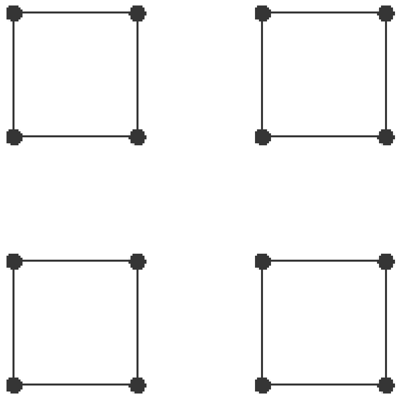
(c) Angelo starts from F and visits E next. He also visits B before he visits D. Find an improved upper bound for Angelo's total travelling time.

(3)

(Total 10 marks)

Q9.

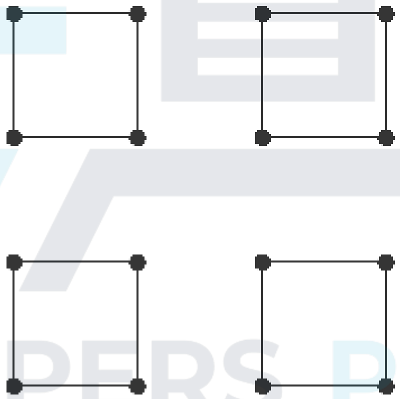
(a) The diagram shows a graph with 16 vertices and 16 edges.



(i) On **Figure 1** below, add the minimum number of edges to make a connected graph.

Figure 1

Connected Graph

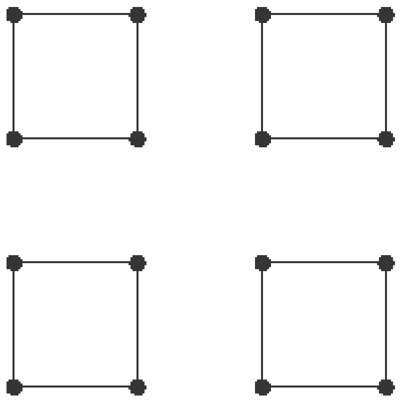


(1)

(ii) On **Figure 2** below, add the minimum number of edges to make the graph Hamiltonian.

Figure 2

Hamiltonian Graph

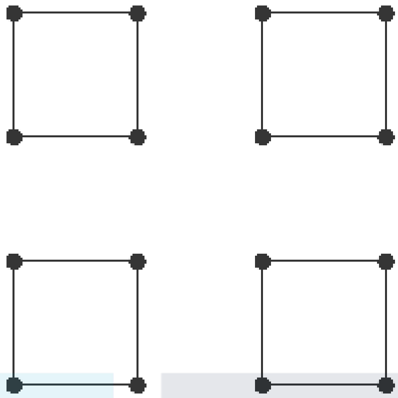


(2)

(iii) On **Figure 3** below, add the minimum number of edges to make the graph Eulerian.

Figure 3

Eulerian Graph



(2)

(b) A complete graph has n vertices and is Eulerian.

(i) State the condition that n must satisfy.

(1)

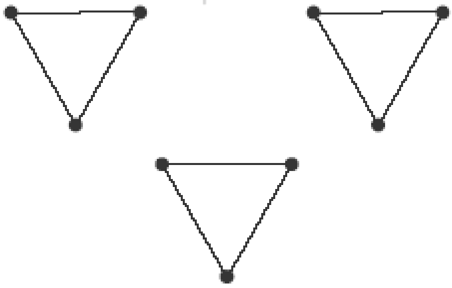
(ii) The number of edges in a Hamiltonian cycle for the graph is the same as the number of edges in an Eulerian trail. State the value of n.

(2)

(Total 8 marks)

Q10.

(a) The diagram shows a graph **G** with 9 vertices and 9 edges.



(i) State the minimum number of edges that need to be added to **G** to make a connected graph. Draw an example of such a graph.

(2)

(ii) State the minimum number of edges that need to be added to **G** to make the graph Hamiltonian. Draw an example of such a graph.

(2)

(iii) State the minimum number of edges that need to be added to **G** to make the graph Eulerian. Draw an example of such a graph.

(2)

(b) A complete graph has n vertices and is Eulerian.

(i) State the condition that n must satisfy.

(1)

(ii) In addition, the number of edges in a Hamiltonian cycle for the graph is the same as the number of edges in an Eulerian trail. State the value of n.

(1)
(Total 8 marks)

Q11.

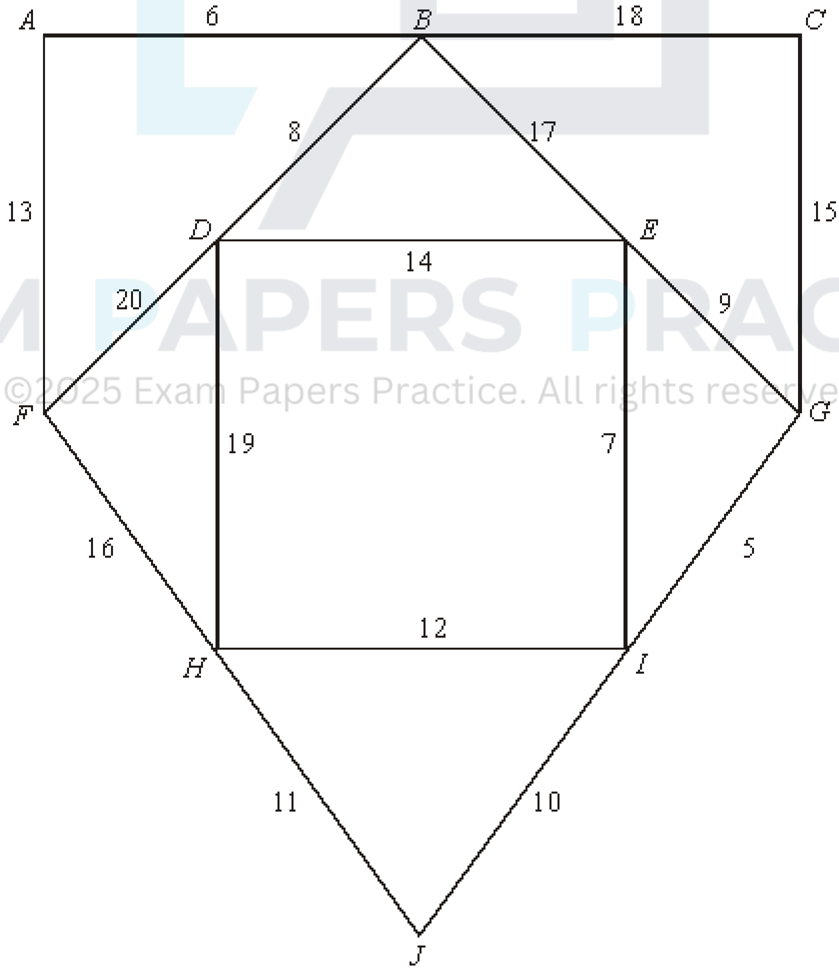
(a) (i) State the number of edges in a minimum spanning tree of a network with 10 vertices.

(1)

(ii) State the number of edges in a minimum spanning tree of a network with n vertices.

(1)

(b) The following network has 10 vertices: A, B, ..., J. The numbers on each edge represent the distances, in miles, between pairs of vertices.



(i) Use Kruskal's algorithm to find the minimum spanning tree for the network.

(5)

(ii) State the length of your spanning tree.

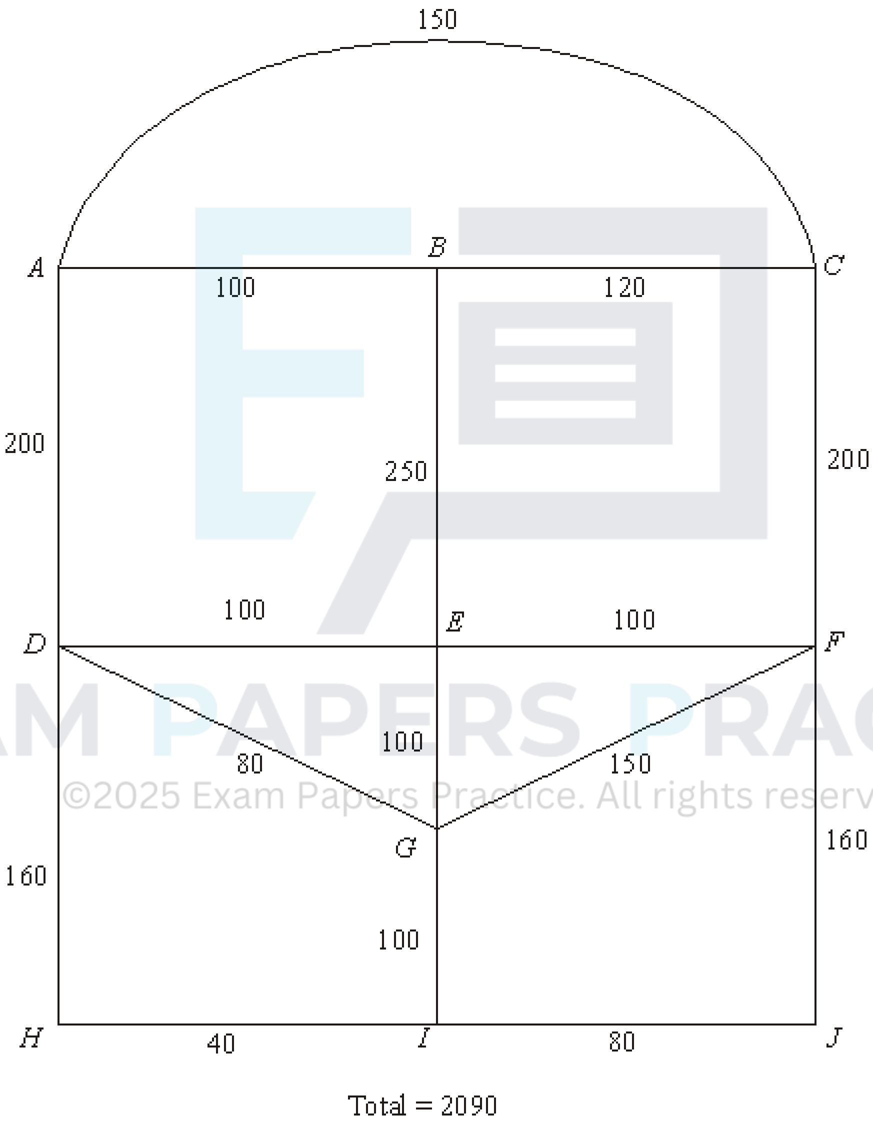
(1)

(iii) Draw your spanning tree.

(2)
(Total 10 marks)

Q12.

Stella is visiting Tijuana on a day trip. The diagram shows the lengths, in metres, of the roads near the bus station.



Stella leaves the bus station at A. She decides to walk along all of the roads at least once before returning to A.

(a) Explain why it is not possible to start from A, travel along each road only once and return to A.

(1)

(b) Find the length of an optimal 'Chinese postman' route around the network, starting and finishing at A.

(5)

(c) At each of the 9 places B, C, ..., J, there is a statue. Find the number of times that

Stella will pass a statue if she follows her optimal route.

(2)
(Total 8 marks)

Q13.

A connected graph **G** has m vertices and n edges.

(a) (i) Write down the number of edges in a minimum spanning tree of **G**.

(1)

(ii) Hence write down an inequality relating m and n.

(2)

(b) The graph **G** contains a Hamiltonian cycle. Write down the number of edges in this cycle.

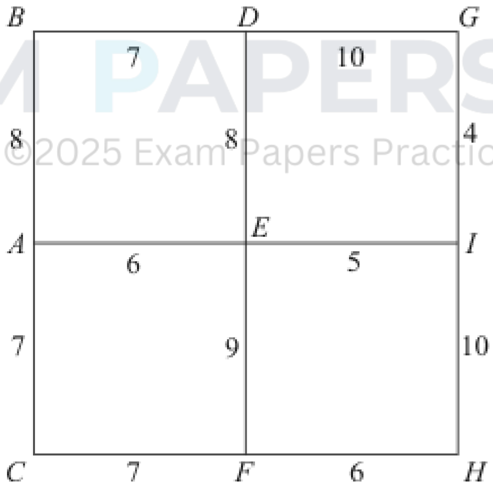
(1)

(c) In the case where **G** is Eulerian, draw a graph of **G** for which $m = 6$ and $n = 12$.

(2)
(Total 6 marks)

Q14.

A local council is responsible for gritting roads.
The diagram shows the length, in miles, of the roads that have to be gritted.



Total length = 87 miles

The gritter is based at A, and must travel along all the roads, at least once, before returning to A.

(a) Explain why it is not possible to start from A and, by travelling along each road only once, return to A.

(1)

(b) Find an optimal 'Chinese postman' route around the network, starting and finishing at A. State the length of your route.

(6)
(Total 7 marks)

Q15.

(a) In the complete graph K_7, every one of the 7 vertices is connected to each of the other 6 vertices by a single edge.

Find or write down:

(i) the number of edges in the graph;

(1)

(ii) the number of edges in a minimum spanning tree;

(1)

(iii) the number of edges in a Hamiltonian cycle.

(1)

(b) (i) Explain why the graph K_7 is Eulerian.

(1)

(ii) Write down the condition for K_n to be Eulerian.

(1)

(c) A connected graph has 6 vertices and 10 edges.

Draw an example of such a graph which is Eulerian.

(2)

(Total 7 marks)

Q16.

A connected graph has five vertices and has arc lengths of

4, 7, 7, 7, 8, 8, 9 and 12 units.

(a) State the minimum length of a minimum spanning tree for any such graph.

(1)

(b) State the minimum length of a Hamiltonian cycle for any such graph.

(1)

(c) State the minimum length of an Eulerian cycle for any such graph.

(1)

(d) In the case when the length of its minimum spanning tree is 26 units, draw a sketch to show a possible graph.

(3)

(Total 6 marks)

Q17.

(a) Draw the graph K_4 .

(2)

(b) (i) Find the total number of edges in K_6 .

(1)

(ii) State the number of edges in a minimum spanning tree of K_6 .

(1)

(iii) Give a reason why K_6 is **not** Eulerian.

(1)

(c) For the graph K_n , state in terms of n:

(i) the total number of edges;

(1)

(ii) the number of edges in a minimum spanning tree;

(1)

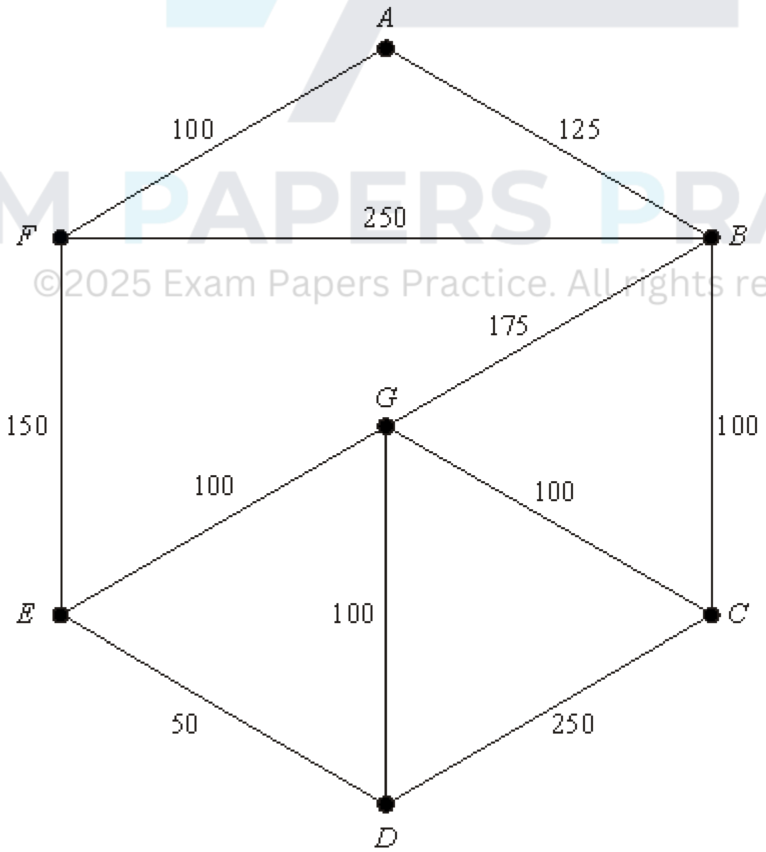
(iii) the condition for K_n to be Eulerian.

(1)

(Total 8 marks)

Q18.

Brendan sells fish from his van. He travels around the streets of an estate. The diagram shows the network of streets he travels along. The number on each arc represents the length, in metres, of that street.



(a) (i) Give a reason why it is **not** possible to start at A, travel along each street once only and return to A.

(1)

(ii) Find the length of an optimal Chinese Postman route around the network, starting and finishing at A. State the number of times each vertex is visited.

(7)

(b) The streets of another estate have 6 odd vertices. Find the number of ways of pairing these odd vertices.

(2)

(Total 10 marks)

Q19.

A graph G has five vertices.

(a) (i) Given that G is connected, state the number of arcs on a minimum spanning tree.

(1)

(ii) Given that G is Eulerian with all vertices having the same degree d, state the minimum value of d.

(1)

(iii) Given that G has a Hamiltonian cycle, state the number of edges in such a cycle.

(1)

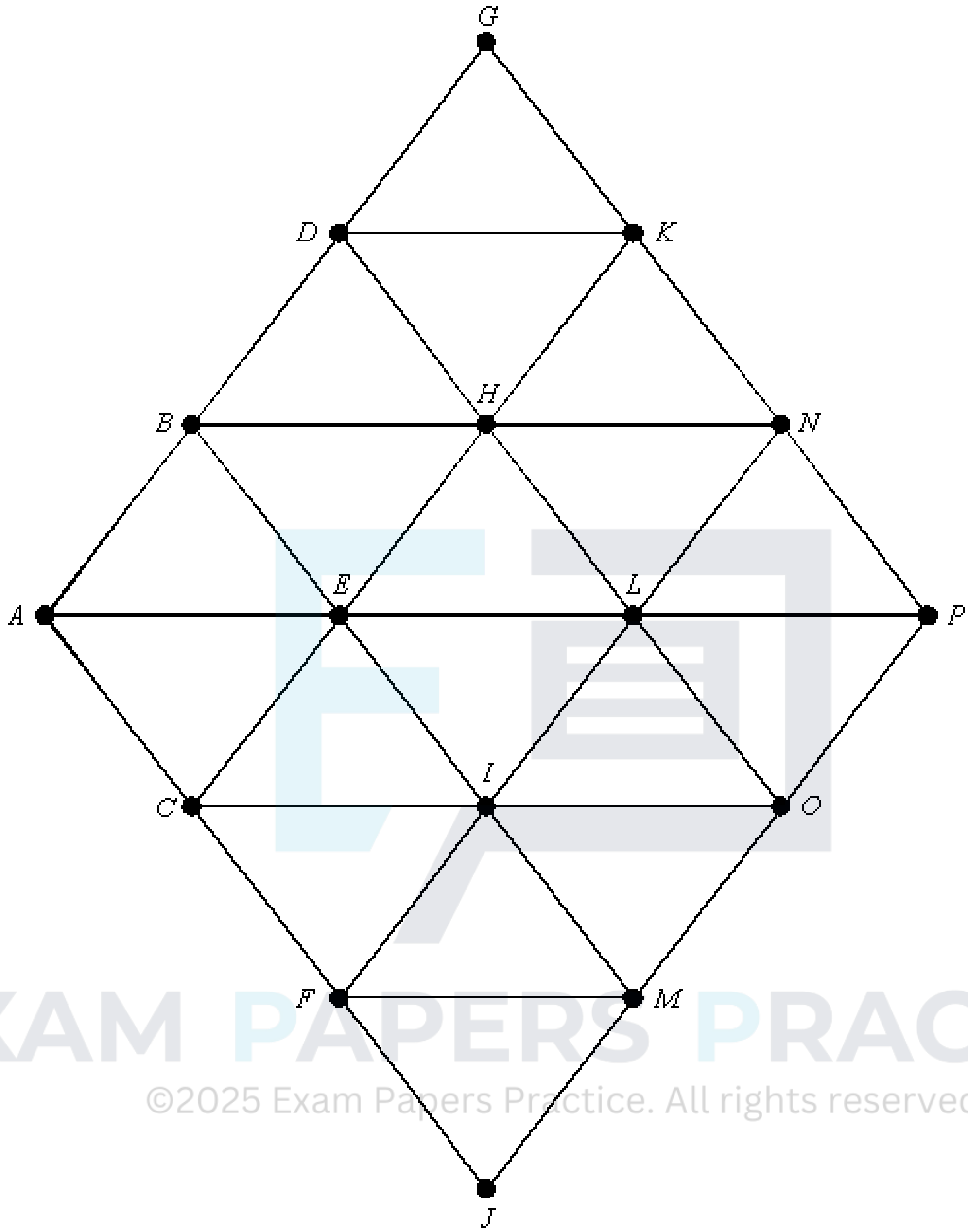
(b) Given that G has all the properties mentioned in part (a), draw a possible graph G.

(1)

(Total 4 marks)

Q20.

The following network has 16 vertices.



(a) Given that the length of each edge is 1 unit, find:

(i) the shortest distance from A to K;

(1)

(ii) the length of a minimum spanning tree.

(2)

(b) (i) Find the length of an optimal Chinese Postman route, starting and finishing at A.

(4)

(ii) For such a route, state the edges that would have to be used twice.

(1)

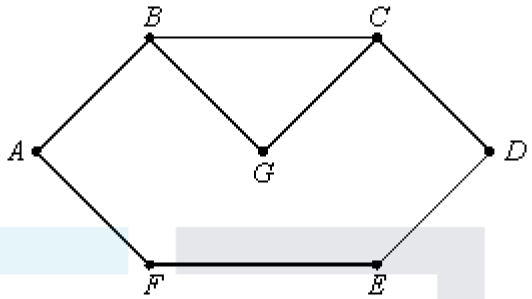
(iii) Given that the edges AE and LP are now removed, find the new length of an optimal Chinese Postman route, starting and finishing at A.

(2)
(Total 10 marks)

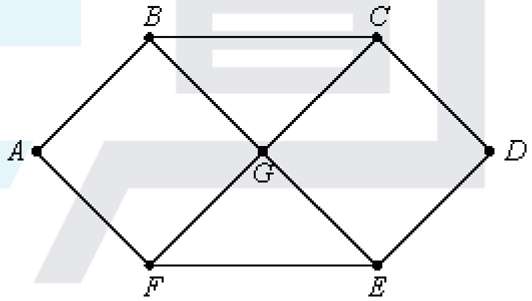
Q21.

The following question refers to the three graphs: **Graph 1**, **Graph 2** and **Graph 3**.

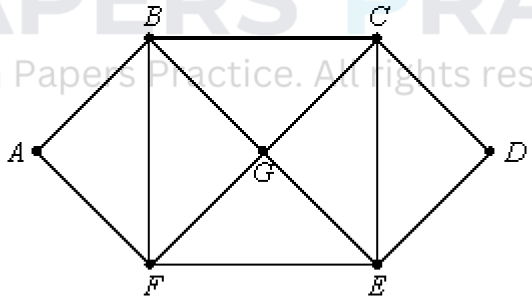
Graph 1



Graph 2



Graph 3



(a) For **each** of the graphs explain whether or not the graph is Eulerian.

(4)

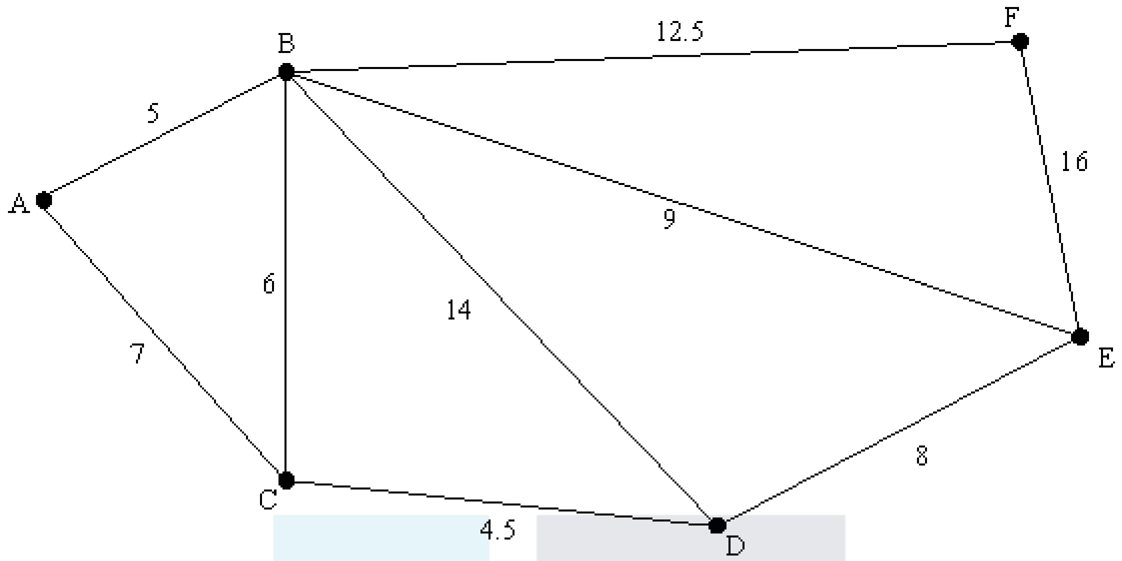
(b) The length of each edge connecting two vertices is 1 unit. Find, for **each** of the graphs, the length of an optimal Chinese postman route, starting and finishing at A.

(4)
(Total 8 marks)

Q22.

A local council is responsible for gritting roads.

(a) The following diagram shows the lengths of roads, in miles, that have to be gritted.



The gritter is based at A and must travel along all the roads, at least once, before returning *to* A.

(i) Explain why it is **not** possible to start from A and, by travelling along each road only once, return to A.

(1)

(ii) Find an optimal 'Chinese postman' route around the network, starting and finishing at A. State the length of your route.

(6)

(b) (i) The connected graph of the roads in the area run by another council has six odd vertices. Find the number of ways of pairing these odd vertices.

(1)

(ii) For a connected graph with n odd vertices, find an expression for the number of ways of pairing these vertices.

(2)

(Total 10 marks)

Q23.

(a) A connected graph has four vertices. State the number of edges in the graph's minimum spanning tree

(1)

(b) A graph has n vertices. The graph is complete, i.e. each vertex is joined to every other vertex by exactly one edge.

(i) State the number of edges in the graph's minimum spanning tree

(1)

(ii) Determine the number of Hamiltonian cycles in the graph.

(2)

(c) A connected graph has four vertices and has arc lengths of

4, 4.5, 5, 6.5, 7, 8 and 9 units.

The length of its minimum spanning tree is 17 units. Draw a sketch to show a possible graph.

(3)
(Total 7 marks)

Q24.

(a) (i) Draw the edges of a graph with the following vertex to vertex table.

	A	B	C	D
A	0	1	2	1
B	1	0	2	1
C	2	2	0	1
D	1	1	1	0

(2)

(ii) Explain why this graph does **not** have an Eulerian cycle.

(1)

(b) A vertex to vertex table is symmetrical about a leading diagonal consisting only of zeros. State the connection between the sum of all the numbers in the table and the number of edges in the corresponding graph.

(1)

(c) State the circumstances in which a graph of a vertex to vertex table is **not** symmetrical about its main diagonal.

(1)
(Total 5 marks)

Q25.

A connected graph G has five vertices. The lengths of the edges are 7, 7, 7, 9, 11, 12, 13, 15 and 16 units, respectively.

(a) A graph is said to be *fully connected* if every vertex is joined at least once to every other vertex. Explain why graph G cannot be fully connected.

(1)

(b) A connected graph G is to be drawn with 5 vertices with lengths of edges as given above

(i) Calculate the least possible length of a minimum spanning tree of graph G.

(1)

(ii) Explain why your answer to part (b)(i) might not apply to graph G.

(1)

(iii) Sketch an example of graph G which has a minimum spanning tree of length 34 units.

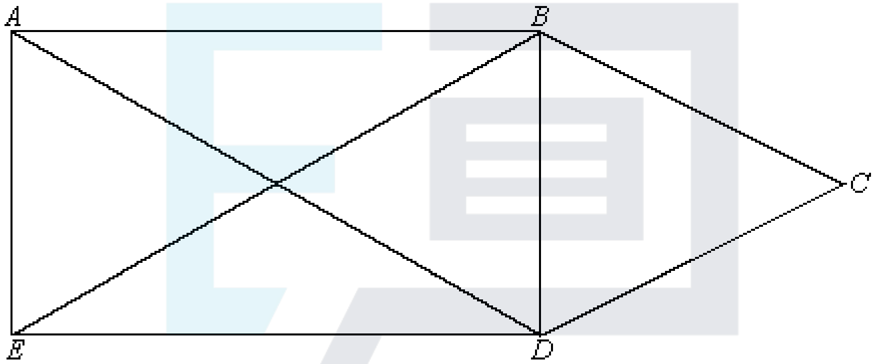
(3)
(Total 6 marks)

Q26.

(a) Explain what is meant by an Eulerian cycle.

(1)

(b) (i) Explain why the following graph cannot have an Eulerian cycle.



(1)

(ii) The graph can be drawn without taking your pen off the paper and without retracing any line. State the vertices from which you could start the drawing.

(1)

(iii) Write down a path that would enable you to draw the graph without taking your pen off the paper and without retracing any line.

(1)
(Total 4 marks)

Q27.

(a) Explain what is meant by a Hamiltonian cycle.

(2)

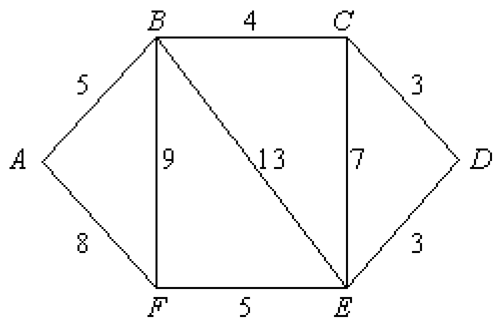
(b) Determine how many different Hamiltonian cycles there are in a fully connected graph with four vertices.

(2)

(c) Write down the number of different Hamiltonian cycles there are in a fully connected graph with n vertices.

(2)
(Total 6 marks)

Q28.



(a) The network represents a road system in which the lengths of the roads are shown in kilometres. The road system has to be cleared of snow by a snow plough which is based at A. The snow plough only needs to travel along each road once in order to clear it. However, when all roads have been cleared it must return to its base at A.

(i) State which roads the snow plough should drive along twice in order to travel the minimum total distance.

(2)

(ii) Hence solve the Chinese postman problem for this network to find a route for the snow plough, starting and finishing at A.

(2)

(iii) The road BE becomes a dual carriageway and must be cleared twice, once in each direction. Redraw the network to take account of this and find, by inspection, a new route for the snow plough.

(3)

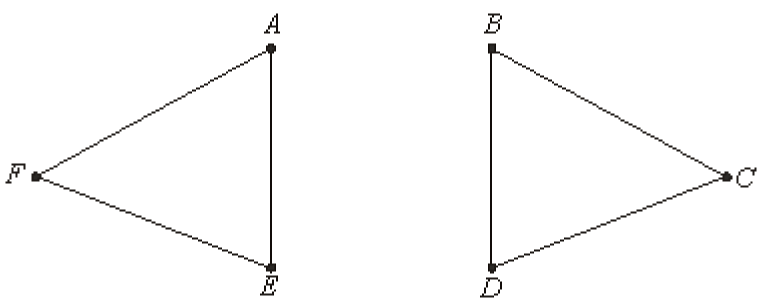
(b) The original network has to be drawn as efficiently as possible by a pen operated by a computer. Explain how this can be done without lifting the pen from the paper and without tracing any of the edges more than once.

(2)

(Total 9 marks)

Q29.

The graph **G** has six vertices A – F and is illustrated below.



(a) What is the minimum number of edges which must be added to **G** in order to make the graph connected?

(1)

(b) What is the minimum number of edges which must be added to **G** in order to make the graph Hamiltonian? State a suitable set of edges.

(2)

(c) (i) State the minimum number of edges which must be added to **G** in order to create a semi-Eulerian graph. Draw such a graph and write down a trail of this graph which uses all its edges.

(3)

(ii) Show that by adding two edges between the same pair of vertices you can create an Eulerian graph.

(1)

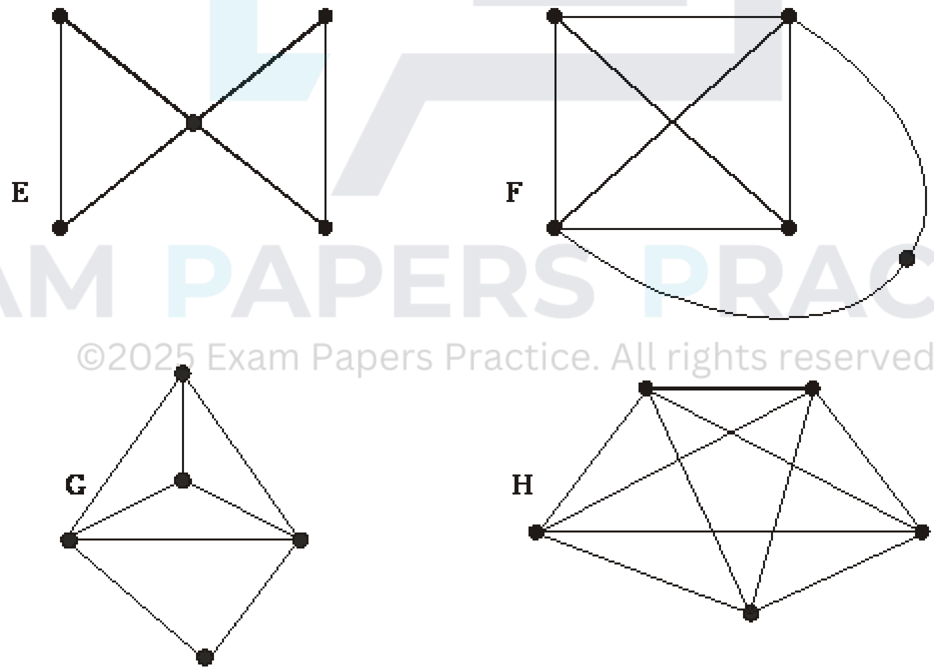
(iii) State the minimum number of edges which must be added to **G** in order to create a **simple** graph which is Eulerian. Draw such a graph.

(3)

(Total 10 marks)

Q30.

The four graphs **E**, **F**, **G** and **H** are illustrated below. They each have five vertices.



(a) State which, if any, of the four graphs are Eulerian.

(2)

(b) State which, if any, of the four graphs are Hamiltonian.

(2)

(c) State which, if any, of the four graphs are planar.

(2)

(d) State which pairs, if any, of the four graphs are isomorphic.

(2)

(Total 8 marks)

Q31.

(a) A simple graph has five vertices and their degrees are d, $d + 1$, $d + 1$, $d + 2$ and $d + 3$.

(i) By considering the sum of the degrees, show that d is odd.

(2)

(ii) Deduce that $d = 1$ and draw a graph with vertices having the given degrees.

(3)

(b) A simple graph has 10 vertices.

(i) Explain why the degree of each vertex must be in the set

$$\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

(1)

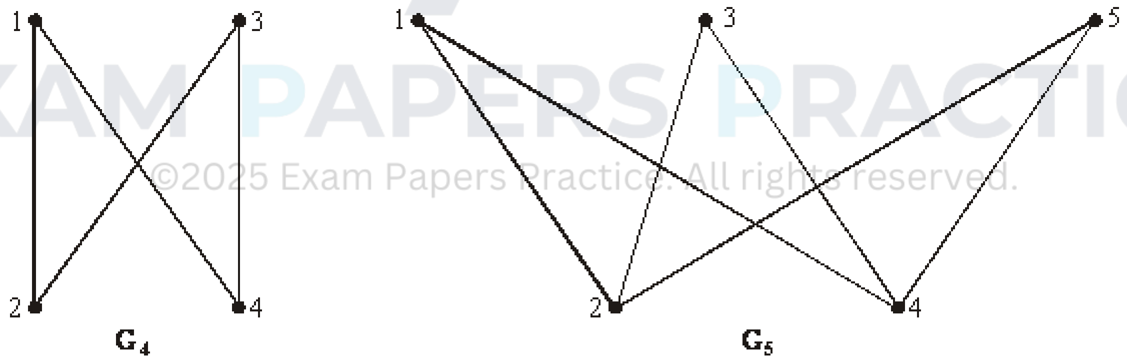
(ii) Show that the degrees of the vertices cannot all be different.

(2)

(Total 8 marks)

Q32.

$\mathbf{G}_n$ is the graph with n vertices labelled 1, 2, 3, . . ., n and with two vertices joined by an edge if the sum of their labels is odd. So, for example, $\mathbf{G}_4$ and $\mathbf{G}_5$ are as drawn below.



(a) Draw $\mathbf{G}_6$.

(2)

(b) (i) Give an example of a Hamiltonian cycle in $\mathbf{G}_6$.

(2)

(ii) For what values of n is $\mathbf{G}_n$ Hamiltonian?

(2)

(c) If n is even what, in terms of n, is the degree of each vertex of $\mathbf{G}_n$?

(1)

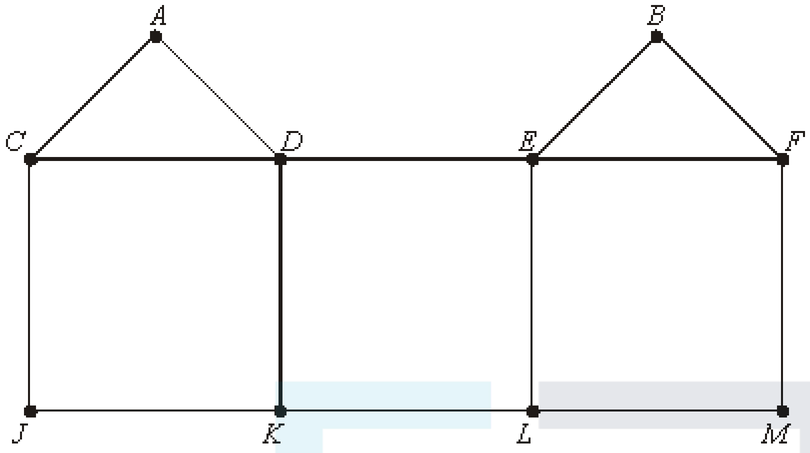
(d) For what values of n is $\mathbf{G}_n$ Eulerian? Justify your answer.

(3)

(Total 10 marks)

Q33.

G is the graph illustrated.



(a) Find the Hamiltonian cycle in **G** which begins CA ....

(2)

(b) Is **G** Eulerian, semi-Eulerian or neither? Give a reason for your answer.

(2)

(c) One edge is removed from **G** to make a semi-Eulerian graph.

(i) State which edge is removed.

(1)

(ii) Explain why the resulting graph is not Hamiltonian.

(2)

(Total 7 marks)

Q34.

A simple graph has six vertices and their degrees are d, $d + 3$, $d + 3$, $d + 3$, $d + 3$ and d + 4.

(i) Explain why d must be either 0 or 1.

(1)

(ii) For which of these values of d will the graph be semi-Eulerian?
Give a reason.

(2)

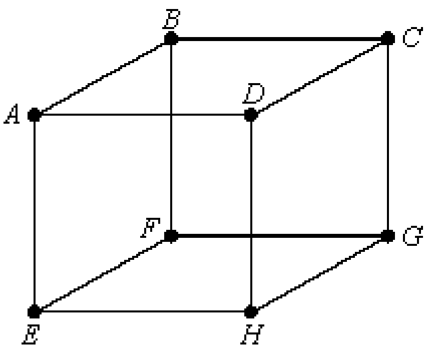
(iii) Show that the graph will never be Hamiltonian.

(2)

(Total 5 marks)

Q35.

A graph with eight vertices $A – H$ is illustrated.

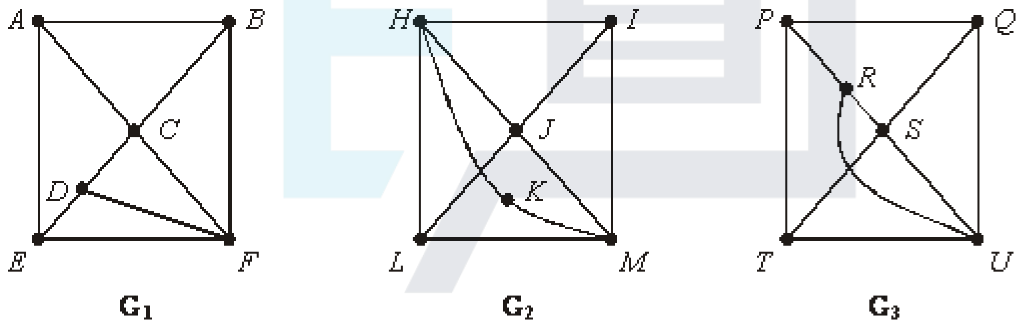


Write down **two** Hamiltonian cycles of the graph which begin ABC....

(Total 3 marks)

Q36.

Let $\mathbf{G}_1$, $\mathbf{G}_2$ and $\mathbf{G}_3$ be the graphs illustrated, each with six vertices.



(a) Explain how you know that none of the graphs is Eulerian.

(1)

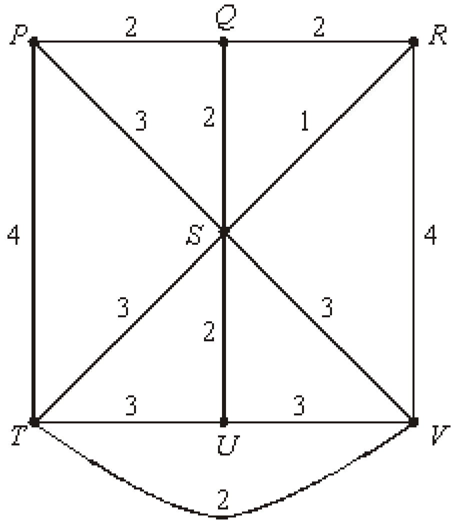
(b) State which of the graphs is semi-Eulerian, and give details of one of its Eulerian trails.

(3)

(Total 4 marks)

Q37.

The network **N** is illustrated below:



(a) Use the Chinese postperson algorithm to find the length of a shortest closed walk in **N** which uses all its edges.

(5)

(b) Apply the nearest neighbour algorithm, starting and finishing at S, to construct a Hamiltonian cycle of **N**. State its length.

(In this case the algorithm works even though the network is not complete.)

(4)

(c) (i) By considering the vertex U, explain why every Hamiltonian cycle of **N** uses at least one edge of length 3.

(ii) By considering **N** with edges PT and RV deleted, explain briefly why each Hamiltonian cycle of **N** uses at least one edge of length 4.

(iii) Show that the Hamiltonian cycle which you constructed in part (b) is the shortest possible in **N**.

(4)

(d) The network **N** represents seven computer centres P, Q, R, S, T, U, V and the direct cable links between them. The number on each edge shows the time, in seconds, that it takes for a piece of information to pass along that link. Whenever a piece of information is received at one centre and then passed on to another there is a further 20 second delay.

(i) A piece of information has to be sent from S to another centre and then passed on to another and then passed on again, until it is eventually sent back to S having been received at least once at each of the other computer centres. Use your earlier answers to find the minimum time that this will take.

(2)

(ii) A technician based at S wants to check every link in the system by sending a piece of information from S to another centre and then on to another and then on again, until it is eventually sent back to S having used **each link** at least once. Use your earlier answers to find the minimum time that this will take.

(3)

(Total 18 marks)

Q38.

A graph **G** has seven vertices labelled 51, 52, 53, 61, 62, 63 and 64. Two vertices are joined by an edge if their two labels use four different digits. (So, for example, 51 is joined to 62, but 51 is not joined to 52 or to 61.)

(a) Draw the graph **G**.

(2)

(b) Explain how you know that **G** is not Eulerian or semi-Eulerian.

(1)

(c) Find an example of a path in **G** from 51 to 61 which uses exactly three edges. How many paths are there from 51 to 61 which use an even number of edges? Explain briefly why each cycle in **G** uses an even number of edges.

(4)

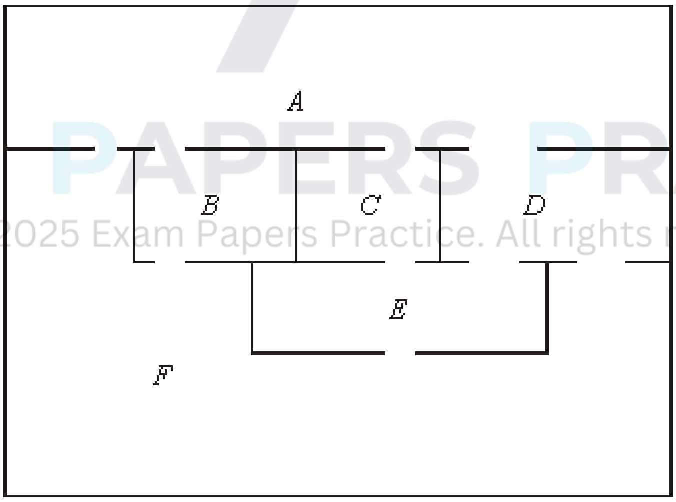
(d) Deduce that **G** is not Hamiltonian.

(2)

(Total 9 marks)

Q39.

The plan shows the rooms $A - F$ of a museum, the gaps in the walls representing doors between the rooms.



(a) Draw a graph with six vertices labelled A, B, C, D, E and F in which two vertices are joined by an edge if there is a door directly from one of the rooms into the other. So for example, there will be an edge DE but there will not be an edge AE.

(3)

(b) State whether your graph is Eulerian, semi-Eulerian or neither. Give a reason for your answer.

(2)

(c) Give an example of a Hamiltonian cycle in your graph.

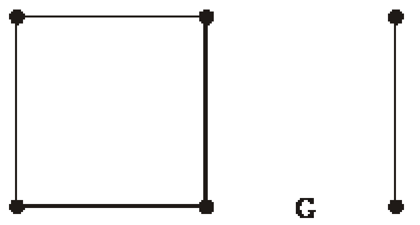
(2)

(d) If you had to design the layout of a museum would you prefer its graph to be Eulerian or Hamiltonian? Give a reason for your answer.

(2)
(Total 9 marks)

Q40.

G is the graph illustrated with 6 vertices and 5 edges:



(a) What is the smallest number of edges which must be added to **G** in order to make a connected graph? Illustrate one such graph,

(1)

(b) What is the smallest number of edges which must be added to **G** in order to make a Hamiltonian graph? Illustrate one such graph, clearly pointing out a Hamiltonian cycle.

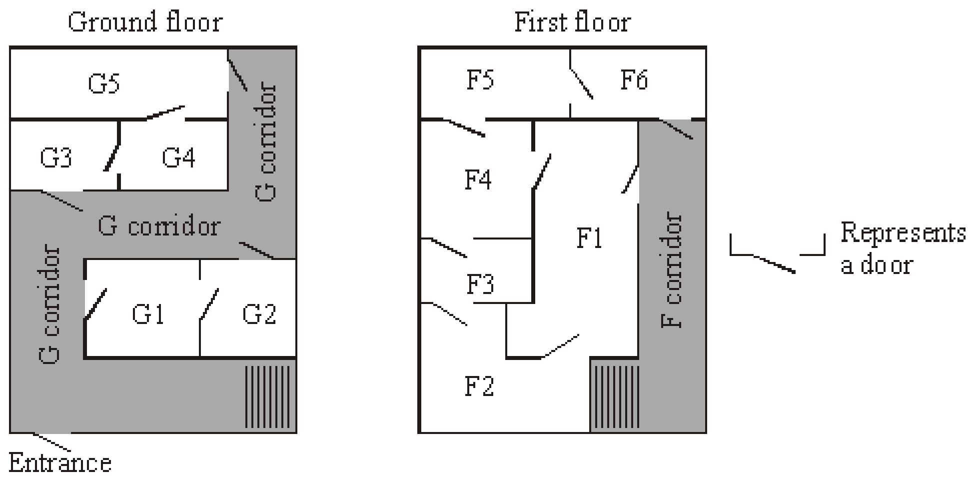
(2)

(c) What is the smallest number of edges which must be added to **G** in order to make an Eulerian graph? Illustrate one such graph.

(2)
(Total 5 marks)

Q41.

The diagram below shows plans of the two floors of a building.



(a) Draw a graph to represent the ground floor with a vertex for each of the five rooms and one vertex for the corridor, and with edges showing which rooms or the corridor are directly connected by a door. Ignore the stairs.

(2)

(b) Similarly, draw a graph for the first floor,

(2)

(c) Show that one of your graphs is Eulerian and that the other is semi-Eulerian, and give an interpretation of this for each floor.

(3)
(Total 7 marks)

Q42.

A graph has eight vertices labelled 1, 2, 3, 4, 5, 6, 7, 8. Two vertices are joined by an edge if their numbers differ by one or two. For example, 2 and 3 are joined, 4 and 6 are joined, but 5 and 8 are not joined.

(a) Draw a picture of the graph.

(3)

(b) Explain how you know that the graph is semi-Eulerian.
Give an example of an Eulerian trail in the graph.

(3)

(c) Give an example of a Hamiltonian cycle in the graph.

(2)

(d) List all paths of length four from vertex 1 to vertex 8.

(3)
(Total 11 marks)

Q43.

A simple graph has vertices A, B, C, D, E, F. The degree of vertex A is m and the degree of each of the other vertices is $2m$, where m is a positive integer.

(a) By considering the sum of the degrees show that m is even, and deduce that $m = 2$.

(2)

(b) Draw such a graph.

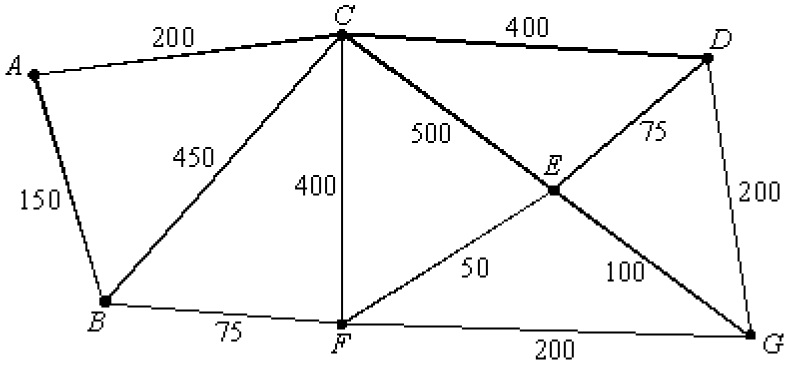
(3)

(c) Calculate the number of graphs which satisfy the given description.

(2)
(Total 7 marks)

Q44.

The graph is a representation of a system of roads. The lengths of the roads are shown in metres.



(a) Use a simple algorithm (not complete enumeration) to find the shortest route from A to G, and give the length of that shortest route.

You must show sufficient detail to demonstrate your use of the algorithm. The answer alone will not gain credit.

(5)

(b) Explain why the graph is not Eulerian.

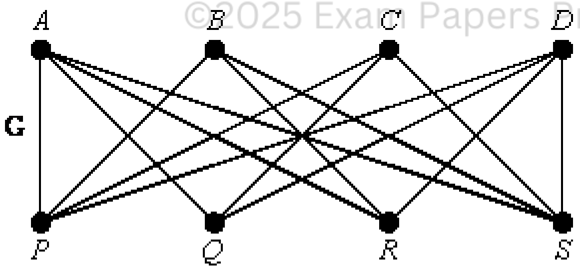
By considering ways of pairing the odd vertices, find a shortest route, starting and finishing at *A*, which traverses each road at least once. State the length of your route.

(6)

(Total 11 marks)

Q45.

The graph **G** shown is the graph $K_{4,4}$.



(a) Explain what is meant by a Hamiltonian cycle, and give an example from **G.**

(3)

(b) How many different Hamiltonian cycles are there in **G?**
(Do not count a cycle taken in reverse order as different from the original.)

(2)

(c) **G** is non-planar.
Explain what this means.
Use Kuratowski's theorem to prove that **G** is non-planar.

(2)

(Total 7 marks)